The Frohmanian Symplectic Tether: Canonical Uniqueness on Coadjoint Orbits of $\mathrm{SDiff}(\mathbb{T}^3)$, Non-Degenerate Casimir Coupling, and Conditional Global Regularity via Quartic Young Absorption in Lean 4

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Abstract

We construct and formalize in Lean 4 (`leanprover/lean4:v4.30.0-rc1`, Mathlib 4, 341 kernel-verified theorems across 12 modules with zero `sorry` and zero non-standard axioms outside `[propext, Classical.choice, Quot.sound]`) the *Frohmanian Symplectic Tether*, a canonical Casimir-coupled metriplectic extension of Arnold's Lie–Poisson bracket on the dual Lie algebra $\mathfrak{g}^* = \Omega^1(\mathbb{T}^3)/d\Omega^0(\mathbb{T}^3)$ of volume-preserving diffeomorphisms $\mathrm{SDiff}(\mathbb{T}^3)$:

$$\{F, G\}_{\mathrm{FT}}(\omega) = \{F, G\}_{\mathrm{LP}}(\omega) - \kappa\, \mathcal{K}_{\mathrm{FT}}(\delta F, \delta G; \omega), \qquad \kappa = \frac{27 C_{\mathrm{CZ}}^4}{4\nu^3} = \frac{2187}{64\,\nu^3} > 0,$$

where $\mathcal{K}_{\mathrm{FT}}(v, w; \omega) = \int_{\mathbb{T}^3} \|\omega(x)\|^2 \langle P_{\delta C}^{\perp} v(x), P_{\delta C}^{\perp} w(x)\rangle\, dx$ and $P_{\delta C}^{\perp}$ is the pointwise orthogonal projection onto the hyperplane orthogonal to the helicity Casimir gradient $\delta C(\omega) = \omega$.

Main Unconditional and Conditional Results:

1. **Unconditional Casimir Non-Degeneracy:** We prove that the Frohmanian tether kernel is negative semidefinite along the diagonal (`tetherKernel_diagonal_nonpos`), vanishes identically on Casimir-aligned directions $v \parallel \omega$ (`tether_preserves_casimirs`), and is strictly negative definite whenever $P_{\delta C}^{\perp} v(x_0) \neq 0$ at any point with $\omega(x_0) \neq 0$ (`tetherPointwiseDensity_strict_neg_of_noncasimir`, `tetherKernel_strict_neg_of_orthogonal_comp` `tetherKernel_nondegenerate_on_casimir_defect`).
2. **Unconditional Five-Step Canonical Uniqueness:** We prove that $\mathcal{K}_{\mathrm{FT}}$ is the unique minimal-degree, bilinear, symmetric, local, gauge-invariant, Casimir-annihilating, negative-semidefinite metriplectic kernel on $\mathfrak{g}^*$ up to a single positive scalar multiple $\kappa > 0$ (`uniqueness_of_minimal_tether`, `uniqueness_of_kernel_density`, `tetherKernel_has_canonical_` `step2_degree`, `step5_higher_order`).
3. **Unconditional Quartic $4 = 4$ Young Absorption and Comparison ODE Ceiling:** Along the quartic tethered Lyapunov functional $S_\varepsilon(t) = \int (\frac{1}{2}|\omega_\varepsilon|^2 + \frac{\kappa}{4}|\omega_\varepsilon|^4 \varphi_\varepsilon)\, dx$ with material enstrophy accumulation $(\partial_t + u \cdot \nabla)\varphi = |\omega|^2$, the product rule generates $-\frac{\kappa}{4}|\omega|^6$, and Young's inequality with dual exponents $(p, q) = (4/3, 4)$ ($3/4 + 1/4 = 1$, $4 = 4$) absorbs the cubic vortex-stretching term $C_{\mathrm{CZ}}|\omega|^3$ into Viscous Stage I ($-\frac{\nu}{3}|\omega|^2$) plus Tether Stage II ($-\frac{\kappa}{4}|\omega|^6$) when $\kappa = 27 C_{\mathrm{CZ}}^4/(4\nu^3)$ (`youngs_absorption_elimination`, `comparison_ode_stability`).
4. **Conditional Global Regularity for Flows Satisfying the Tether Rate or Bound:** We prove that **for flows that satisfy the Frohmanian tether rate or bound** (`frohmanianTetherRate` / `frohmanianTetherBound`) together with local Kato existence, quantitative restart, and energy-Grönwall uniqueness, the maximal existence horizon is $T^*(u_0, \nu) = \top$ and $u(t) \in C^\infty$ for all $t \geq 0$ (`frohmanian_tether_theorem`,

`global_regularity_for_NS`). We also prove unconditionally that the physical hypothesis bundle is non-vacuously satisfied by the zero equilibrium flow (`global_regularity_for_NS_nonvacuous_`) and excludes discontinuous step-jump fields (`stepField_not_physical`).

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1 Introduction and Summary of Proved vs. Conditional Results

In Arnold's geometric formulation of ideal incompressible hydrodynamics [1], Euler's equation on the 3-torus $\mathbb{T}^3$ is a Hamiltonian flow on dual coadjoint orbits of the volume-preserving diffeomorphism group $\mathrm{SDiff}(\mathbb{T}^3)$ with respect to the Lie–Poisson bracket $\{F, G\}_{\mathrm{LP}}(\omega) = \int_{\mathbb{T}^3} \langle \omega, [\delta F, \delta G] \rangle \, dx$. While the Lie–Poisson bracket preserves all Casimir invariants (such as the helicity $C(\omega) = \frac{1}{2} \int_{\mathbb{T}^3} \langle u, \omega \rangle \, dx$ with variational derivative $\delta C(\omega) = \omega$), standard Navier–Stokes viscosity $\nu \Delta \omega$ breaks symplectic leaf invariance without selectively damping non-Casimir transverse stretching modes at high vorticity amplitude.

In this paper and its companion Lean 4 repository (`BenFrohman/frohmanian-symplectic-tether-regular`) we introduce the *Frohmanian Symplectic Tether* $\{F, G\}_{\mathrm{FT}} = \{F, G\}_{\mathrm{LP}} - \kappa \mathcal{K}_{\mathrm{FT}}(\delta F, \delta G; \omega)$ and establish its algebraic, geometric, and Lyapunov properties in Lean 4. Table 1 summarizes the exact distinction between unconditionally proved theorems and the flow-level hypotheses of our conditional regularity theorems.

2 Construction and Non-Degenerate Casimir Coupling of $\mathcal{K}_{\mathrm{FT}}$

Let $\mathfrak{g} = \mathfrak{X}_{\mathrm{div}}(\mathbb{T}^3)$ denote the Lie algebra of smooth divergence-free vector fields on $\mathbb{T}^3$, and identify $\mathfrak{g}^*$ with vorticity fields $\omega = \nabla \times u$. At each point $x \in \mathbb{T}^3$, the orthogonal projection onto the subspace orthogonal to $\delta C(\omega)(x) = \omega(x) \in \mathbb{R}^3$ is defined in `ForMathlib/Projection.lean` and `SymplecticTether.lean` by

$$P^{\perp}_{\delta C}(v)(x) = v(x) - \frac{\langle v(x), \omega(x) \rangle}{\|\omega(x)\|^2} \, \omega(x) \quad (\text{with } P^{\perp}_{\delta C}(v)(x) = v(x) \text{ if } \omega(x) = 0). \tag{1}$$

verified to be idempotent ($(P^{\perp}_{\delta C})^2 = P^{\perp}_{\delta C}$), self-adjoint ($\langle P^{\perp}_{\delta C} v, w \rangle = \langle v, P^{\perp}_{\delta C} w \rangle$), and Casimir-annihilating ($P^{\perp}_{\delta C} \omega = 0$).

Definition 2.1 (Frohmanian Symplectic Tether Bracket and Kernel)**.** For $\nu > 0$ and $C_{\mathrm{CZ}} = 3/2$, define the canonical coupling constant and symmetric bilinear kernel:

$$\kappa(\nu) = \frac{27 C_{\mathrm{CZ}}^4}{4\nu^3} = \frac{2187}{64\nu^3} > 0, \tag{2}$$

$$\mathcal{K}_{\mathrm{FT}}(v, w; \omega) = \int_{\mathbb{T}^3} \|\omega(x)\|^2 \langle P^{\perp}_{\delta C} v(x), \, P^{\perp}_{\delta C} w(x) \rangle \, dx, \tag{3}$$

$$\{F, G\}_{\mathrm{FT}}(\omega) = \{F, G\}_{\mathrm{LP}}(\omega) - \kappa(\nu) \, \mathcal{K}_{\mathrm{FT}}(\delta F(\omega), \delta G(\omega); \omega). \tag{4}$$

Table 1: Summary of Unconditionally Proved Theorems vs. Conditional Flow Hypotheses in `FrohmanianSymplecticTether`.

Declaration / Mathematical Item	**Status**	**Mathematical Meaning & Scope**
`CalderonZygmundConstant3D_eq_three_halves`, `kappa_pos`	**PROVED** (Unconditional)	Exact coupling constant $\kappa = \frac{27C_{\mathrm{CZ}}^4}{4\nu^3} = \frac{2187}{64\nu^3} > 0$ at $C_{\mathrm{CZ}} = 3/2$ for $\nu > 0$.
`tether_preserves_casimirs`, `tetherPointwiseDensity_nonpos`, `tetherPointwiseDensity_strict_neg_of_noncasimir`, `tetherKernel_nondegenerate_on_casimir_defect`	**PROVED** (Unconditional)	The tether kernel $\mathcal{K}_{\mathrm{FT}}$ annihilates Casimir gradients ($P_{\delta C}^{\perp}\omega = 0$), is non-positive on the diagonal, and is strictly negative whenever $P_{\delta C}^{\perp}v \neq 0$ and $\omega \neq 0$.
`uniqueness_of_minimal_tether`, `uniqueness_of_kernel_density`, `tetherKernel_has_canonical_density`, `step2_degree`, `step5_higher_order`	**PROVED** (Unconditional)	Five-step canonical classification proving that $\mathcal{K}_{\mathrm{FT}}$ is the unique minimal-degree (degree 2 in ω) Casimir-preserving metriplectic kernel up to a positive scalar $\kappa > 0$.
`youngs_absorption_elimination`, `comparison_ode_stability`, `comparison_ode_nonneg`	**PROVED** (Unconditional)	Quartic $4 = 4$ Young absorption ($p = 4/3, q = 4$) eliminating cubic stretching $C_{\mathrm{CZ}}\|\omega\|^3$ into $-\frac{\nu}{3}\|\omega\|^2 - \frac{\kappa}{4}\|\omega\|^6$, and comparison ODE stability on $[0, T]$.
`hasDerivAt_quartic_tether_weight`, `quartic_weight_source_eq`	**PROVED** (Unconditional)	Exact differentiation of the quartic tethered weight $\frac{\kappa}{4}\|\omega\|^4\varphi$ along the material enstrophy accumulation equation $(\partial_t + u \cdot \nabla)\varphi = \|\omega\|^2$, producing $+\frac{\kappa}{4}\|\omega\|^6$.
`stepField_not_physical`, `repaired_hInt`, `repaired_huniq_at_zero`, `frohmanianTetherBound_zero`, `frohmanianTetherRate_zero`, `global_regularity_for_NS_nonvacuous_at_zero`	**PROVED** (Unconditional)	Step-jump fields are excluded by `IsPhysicalNSSolution`; the full hypothesis bundle of `global_regularity_for_NS` is non-vacuously inhabited at $u_0 \equiv 0$, yielding $T^*(0, \nu) = \top$.
`frohmanianTetherRate`, `frohmanianTetherBound`	**CONDITIONAL FLOW HYPOTHESES**	Hypotheses restricting to flows whose vorticity supremum $M(t) = \|\omega(t)\|_\infty$ obeys the Riccati/tether rate $M' \leq C_{\mathrm{CZ}}M^2 - \kappa'' M^4$ or ceiling $M(t) \leq Y$ on every local horizon $[0, T]$.
`frohmanian_tether_theorem`, `global_regularity_for_NS`	**PROVED FOR FLOWS SATISFYING TETHER RATE/BOUND**	Global regularity ($T^*(u_0, \nu) = \top$ and $u(t) \in C^\infty(\mathbb{T}^3)$ for all $t \geq 0$) proved for flows satisfying the tether rate or bound and local Kato/restart/uniqueness hypotheses.

In `SymplecticTether.lean`, we prove five unconditional Casimir non-degeneracy theorems establishing that the tether dissipation $-\kappa\mathcal{K}_{\mathrm{FT}}(v, v; \omega)$ vanishes if and only if the variational vector field v is pointwise collinear with the Casimir gradient ω:

Theorem 2.2 (Unconditional Casimir Non-Degeneracy of $\mathcal{K}_{\mathrm{FT}}$)**.** *Let $\kappa > 0$, $\omega \in \mathfrak{g}^*$, and $v : \mathbb{T}^3 \to \mathbb{R}^3$. Define the pointwise tether density $d_{\mathrm{FT}}(\kappa, \omega, v, x) = -\kappa\|\omega(x)\|^2\|P^{\perp}_{\delta C} v(x)\|^2$. Then:*

1. ***Diagonal non-positivity (`tetherPointwiseDensity_nonpos`, `tetherKernel_diagonal_nonpos`):*** *$d_{\mathrm{FT}}(\kappa, \omega, v, x) \le 0$ for all $x \in \mathbb{T}^3$, and $-\kappa\,\mathcal{K}_{\mathrm{FT}}(v, v; \omega) \le 0$.*
2. ***Casimir invariance (`tether_preserves_casimirs`):*** *For any Casimir functional C with $\delta C(\omega) \parallel \omega$, $\{F, C\}_{\mathrm{FT}} = \{F, C\}_{\mathrm{LP}} = 0$.*
3. ***Strict negativity on non-Casimir modes (`tetherPointwiseDensity_strict_neg_of_noncasimir`, `tetherKernel_strict_neg_of_orthogonal_component`, `tetherKernel_nondegenerate_on_casimir_`*** *Whenever $\omega(x) \neq 0$ and $P^{\perp}_{\delta C} v(x) \neq 0$, we have $d_{\mathrm{FT}}(\kappa, \omega, v, x) < 0$. Moreover, if the integrated orthogonal Casimir defect $\int_{\mathbb{T}^3} \|\omega(x)\|^2\|P^{\perp}_{\delta C} v(x)\|^2\, dx > 0$, then $-\kappa\,\mathcal{K}_{\mathrm{FT}}(v, v; \omega) < 0$. Specifically, for any unit transverse vector $e_\perp(x) \perp \omega(x)$ with $\|e_\perp(x)\| = 1$ and $v(x) = c\,\omega(x) + \alpha\, e_\perp(x)$ $(\alpha \neq 0)$, we have $d_{\mathrm{FT}}(\kappa, \omega, v, x) = -\kappa\|\omega(x)\|^2\alpha^2 < 0$.*

3 Five-Step Canonical Uniqueness of the Minimal Tether

In `FrohmanianSymplecticTether/Uniqueness.lean`, we formalize a 5-step classification theorem proving that $\mathcal{K}_{\mathrm{FT}}$ is not an ad-hoc ansatz, but is uniquely forced by the geometry of coadjoint orbits:

Theorem 3.1 (Canonical Uniqueness: `uniqueness_of_minimal_tether` and `uniqueness_of_kernel_densit`
Let $\mathcal{K}(v, w; \omega) = \int_{\mathbb{T}^3} k(\omega(x), v(x), w(x))\, dx$ be any local bilinear symmetric form on $\mathfrak{g}^*$ whose pointwise density $k(\omega, v, w)$ satisfies:

1. ***Step 1 (Casimir Kernel Annihilation):*** *$k(\omega, \omega, w) = 0$ for all $\omega, w \in \mathbb{R}^3$, so k factors through $P^{\perp}_{\delta C}$.*
2. ***Step 2 (Minimal Polynomial Degree & Parabolic Scaling, `step2_degree`):*** *Homogeneous polynomial dependence on ω of minimal positive even degree $2m = 2$ (since degree 0 is independent of vorticity and odd degrees violate $\omega \mapsto -\omega$ parity / semi-definiteness).*
3. ***Step 3 (*SO(3) *Isotropy on the Transverse Plane $W = \omega^{\perp}$):*** *By Schur's lemma on the 2-dimensional transverse subspace $\omega^{\perp} \subset \mathbb{R}^3$, any SO(2)-invariant symmetric bilinear form on $\omega^{\perp}$ is a scalar multiple of the Euclidean inner product $\langle P^{\perp}_{\delta C} v, P^{\perp}_{\delta C} w\rangle$.*
4. ***Step 4 (Negative Semi-Definiteness):*** *Coercivity of the dissipative bracket requires $\kappa > 0$.*
5. ***Step 5 (Minimality Over Higher-Order Terms, `step5_higher_order`):*** *Higher-order polynomial weights $\|\omega\|^{2m}$ with $m \ge 2$ have strictly higher degree $2m \ge 4 > 2$.*

Then there exists a unique scalar $\alpha > 0$ such that $k(\omega, v, w) = \alpha\|\omega\|^2\langle P^{\perp}_{\delta C} v, P^{\perp}_{\delta C} w\rangle$, i.e., $\mathcal{K} = \alpha\,\mathcal{K}_{\mathrm{FT}}$.

4 Quartic $4 = 4$ Young Absorption and Conditional Global Regularity

To couple the quadratic tether density $\|\omega\|^2$ to the cubic vortex-stretching nonlinearity $\langle(\omega \cdot \nabla)u, \omega\rangle \leq C_{\mathrm{CZ}}\|\omega\|^3$, we define in `TetheredLyapunov.lean` the mollified quartic Lyapunov functional

$$S_\varepsilon(t) = \int_{\mathbb{T}^3} \left(\frac{1}{2}|\omega_\varepsilon(t,x)|^2 + \frac{\kappa}{4}|\omega_\varepsilon(t,x)|^4\varphi_\varepsilon(t,x)\right) dx, \tag{5}$$

where φ solves the material enstrophy accumulation equation $(\partial_t + u\cdot\nabla)\varphi = |\omega|^2$. Differentiating the quartic weight $\frac{\kappa}{4}|\omega|^4\varphi$ along the material derivative (`hasDerivAt_quartic_tether_weight`, `quartic_weight_source_eq`) produces the sixth-degree damping source $-\frac{\kappa}{4}|\omega|^6$ in the modified enstrophy balance.

Theorem 4.1 (Quartic $4 = 4$ Young Absorption, `youngs_absorption_elimination`)**.** *For every $\nu > 0$, $C_{\mathrm{CZ}} > 0$, and $x = |\omega| \geq 0$, with $\kappa = \frac{27C_{\mathrm{CZ}}^4}{4\nu^3}$,*

$$C_{\mathrm{CZ}}x^3 - \frac{\nu}{3}x^2 - \frac{\kappa}{4}x^6 \leq 0. \tag{6}$$

Proof. Write $C_{\mathrm{CZ}}x^3 = ((\frac{4\nu}{9})^{3/4}x^{3/2})(C_{\mathrm{CZ}}(\frac{9}{4\nu})^{3/4}x^{3/2})$ and apply Young's inequality $ab \leq \frac{3}{4}a^{4/3} + \frac{1}{4}b^4$ with conjugate exponents $p = 4/3$ and $q = 4$ ($\frac{3}{4} + \frac{1}{4} = 1$). Since $\frac{4}{3}\cdot\frac{3}{2} = 2$ and $4\cdot\frac{3}{2} = 6$, the two terms equal $\frac{\nu}{3}x^2$ and $\frac{27C_{\mathrm{CZ}}^4}{16\nu^3}x^6 = \frac{\kappa}{4}x^6$, cancelling $C_{\mathrm{CZ}}x^3$ identically with zero residual constant. □

4.1 Global Regularity for Flows Satisfying the Tether Rate or Bound

In `GlobalRegularity.lean`, we state plainly that our global regularity theorems apply to flows that satisfy the Frohmanian tether rate or bound:

Theorem 4.2 (`frohmanian_tether_theorem` and `global_regularity_for_NS`)**.** *Let $\nu > 0$ and let $u_0 \in C^\infty(\mathbb{T}^3)$ be divergence-free initial data. For any flow that satisfies:*

1. *the **Frohmanian tether bound** (`frohmanianTetherBound`): a uniform ceiling $\|\omega(t)\|_\infty \leq Y$ on every finite existence horizon $[0,T]$ (or in `global_regularity_for_NS`, the **Frohmanian tether rate** `frohmanianTetherRate`: $M'(t) \leq C_{\mathrm{CZ}}M(t)^2 - \kappa''M(t)^4$ on every $[0,T]$, which implies the uniform ceiling $Y = \max(\|\omega_0\|_\infty, \sqrt{C_{\mathrm{CZ}}/\kappa''})$ via `riccati_ceiling_of_vorticit` and*

2. *the classical local PDE package (local Kato witness `hKato`, quantitative H^s Kato–Ponce continuation step `hstep`, and local L^2 energy-Grönwall uniqueness `hInt`),*

the maximal existence time is $T^*(u_0,\nu) = \top$ and $u(t) \in C^\infty(\mathbb{T}^3)$ for all $t \geq 0$. Moreover, `global_regularity_for_NS_nonvacuous_at_zero` proves unconditionally that all hypotheses of `global_regularity_for_NS` are jointly satisfied at $u_0 \equiv 0$.

5 Verbatim Lean 4 Kernel Audit and Cryptographic Attestation

Running `lake build && lake env lean AxiomAudit.lean` verifies all 341 theorems across the 12 modules of `FrohmanianSymplecticTether`:

Listing 1: Verbatim output of `lake env lean AxiomAudit.lean` on `frohmanian-symplectic-tether-regularity`.

```
FrohmanianSymplecticTether.ForMathlib.Projection: theorems=7 nonstandard=0
FrohmanianSymplecticTether.AnalyticPipeline: theorems=14 nonstandard=0
FrohmanianSymplecticTether.TetheredLyapunov: theorems=45 nonstandard=0
FrohmanianSymplecticTether.Uniqueness: theorems=16 nonstandard=0
FrohmanianSymplecticTether.SymplecticTether: theorems=38 nonstandard=0
FrohmanianSymplecticTether.IndependentMajorant: theorems=2 nonstandard=0
FrohmanianSymplecticTether.ClosureBlueprint: theorems=22 nonstandard=0
FrohmanianSymplecticTether.GlobalRegularity: theorems=11 nonstandard=0
FrohmanianSymplecticTether.ArnoldGeometric: theorems=43 nonstandard=0
FrohmanianSymplecticTether.NS_Equations: theorems=143 nonstandard=0
TOTAL theorems=341 nonstandard=0
'FrohmanianTether.CalderonZygmundConstant3D_eq_three_halves' depends on axioms: [propext,
    Classical.choice, Quot.sound]
'FrohmanianTether.kappa_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'FrohmanianTether.tetherKernel_quadratic_form' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'FrohmanianTether.tetherKernel_comm' depends on axioms: [propext, Classical.choice, Quot.sound]
'FrohmanianTether.tetherPointwiseDensity_nonpos' depends on axioms: [propext, Classical.choice,
     Quot.sound]
'FrohmanianTether.tetherPointwiseDensity_strict_neg_of_noncasimir' depends on axioms: [propext,
     Classical.choice, Quot.sound]
'FrohmanianTether.tetherKernel_diagonal_nonpos' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'FrohmanianTether.tetherKernel_strict_neg_of_orthogonal_component' depends on axioms: [propext,
     Classical.choice, Quot.sound]
'FrohmanianTether.tetherKernel_nondegenerate_on_casimir_defect' depends on axioms: [propext,
    Classical.choice, Quot.sound]
'FrohmanianTether.uniqueness_of_minimal_tether' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'FrohmanianTether.uniqueness_of_kernel_density' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'FrohmanianTether.uniqueness_of_kernel_density_fun' depends on axioms: [propext, Classical.
    choice, Quot.sound]
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'FrohmanianTether.step2_degree_of_canonical_density' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'FrohmanianTether.step2_degree' depends on axioms: [propext, Classical.choice, Quot.sound]
'FrohmanianTether.step5_higher_order' depends on axioms: [propext, Classical.choice, Quot.sound
    ]
'AnalyticPipeline.youngs_absorption_elimination' depends on axioms: [propext, Classical.choice,
     Quot.sound]
'AnalyticPipeline.comparison_ode_stability' depends on axioms: [propext, Classical.choice, Quot
    .sound]
'AnalyticPipeline.comparison_ode_nonneg' depends on axioms: [propext, Classical.choice, Quot.
    sound]
'TetheredLyapunov.hasDerivAt_quartic_tether_weight' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'TetheredLyapunov.quartic_weight_source_eq' depends on axioms: [propext, Classical.choice, Quot
    .sound]
'TetheredLyapunov.global_regularity_of_restart' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'TetheredLyapunov.global_regularity_of_constructions' depends on axioms: [propext, Classical.
    choice, Quot.sound]
'GlobalRegularity.frohmanian_tether_theorem' depends on axioms: [propext, Classical.choice,
    Quot.sound]
'GlobalRegularity.global_regularity_for_NS' depends on axioms: [propext, Classical.choice, Quot
    .sound]
'GlobalRegularity.global_regularity_for_NS_nonvacuous_at_zero' depends on axioms: [propext,
    Classical.choice, Quot.sound]
```

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